Analysis of Skiing and Snowboarding Injuries Using NEISS Data (2016–2024)

Individual Programming Project

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COMP 641

Why Study Ski & Snowboard Injuries?

→ Scale of Participation

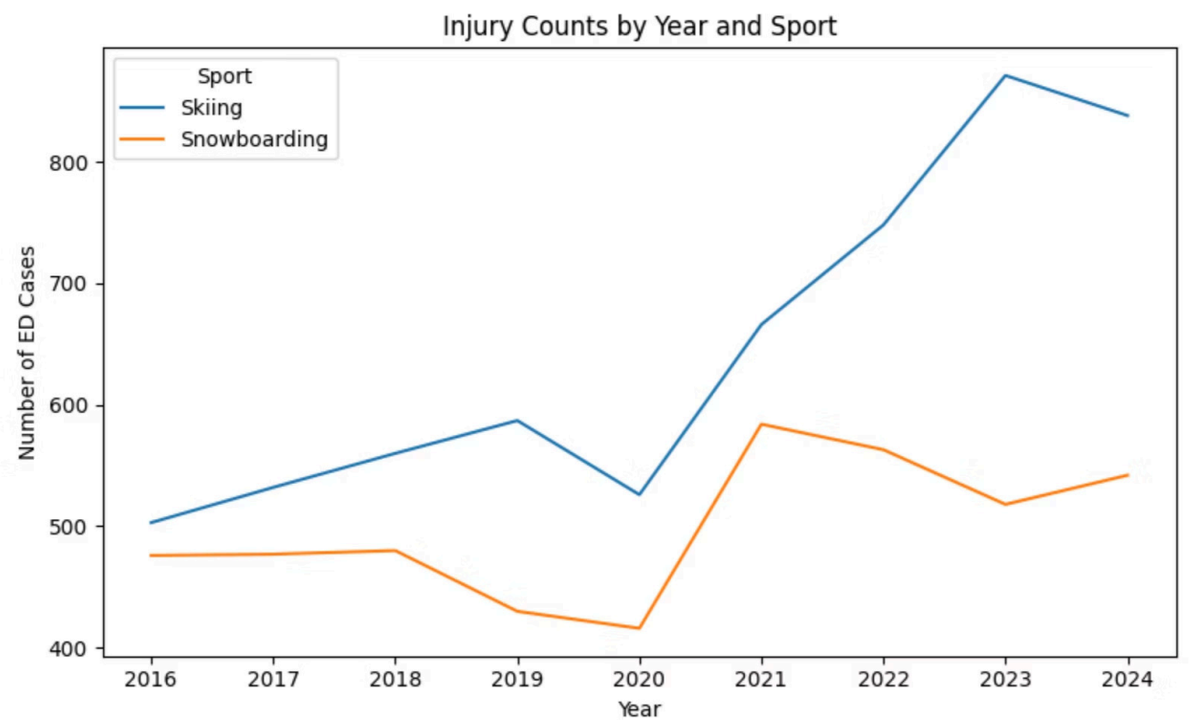
Millions engage in alpine skiing and snowboarding annually, making winter sports among the most widely practiced recreational activities in the U.S.

→ Persistent Safety Risk

Despite equipment advances, these injuries remain a consistent concern: sprains, fractures, and head trauma account for significant ED visits each season.

→ Public Health Value

Systematic surveillance enables identification of high-risk demographics and informs evidence-based prevention strategies and policy.



Injury counts increased through the pre-pandemic years, declined in 2020, and rose again afterward. The 2020 decline likely reflects reduced participation during COVID-19 rather than a true change in injury risk.

Data Source: NEISS Injury Surveillance System

About NEISS

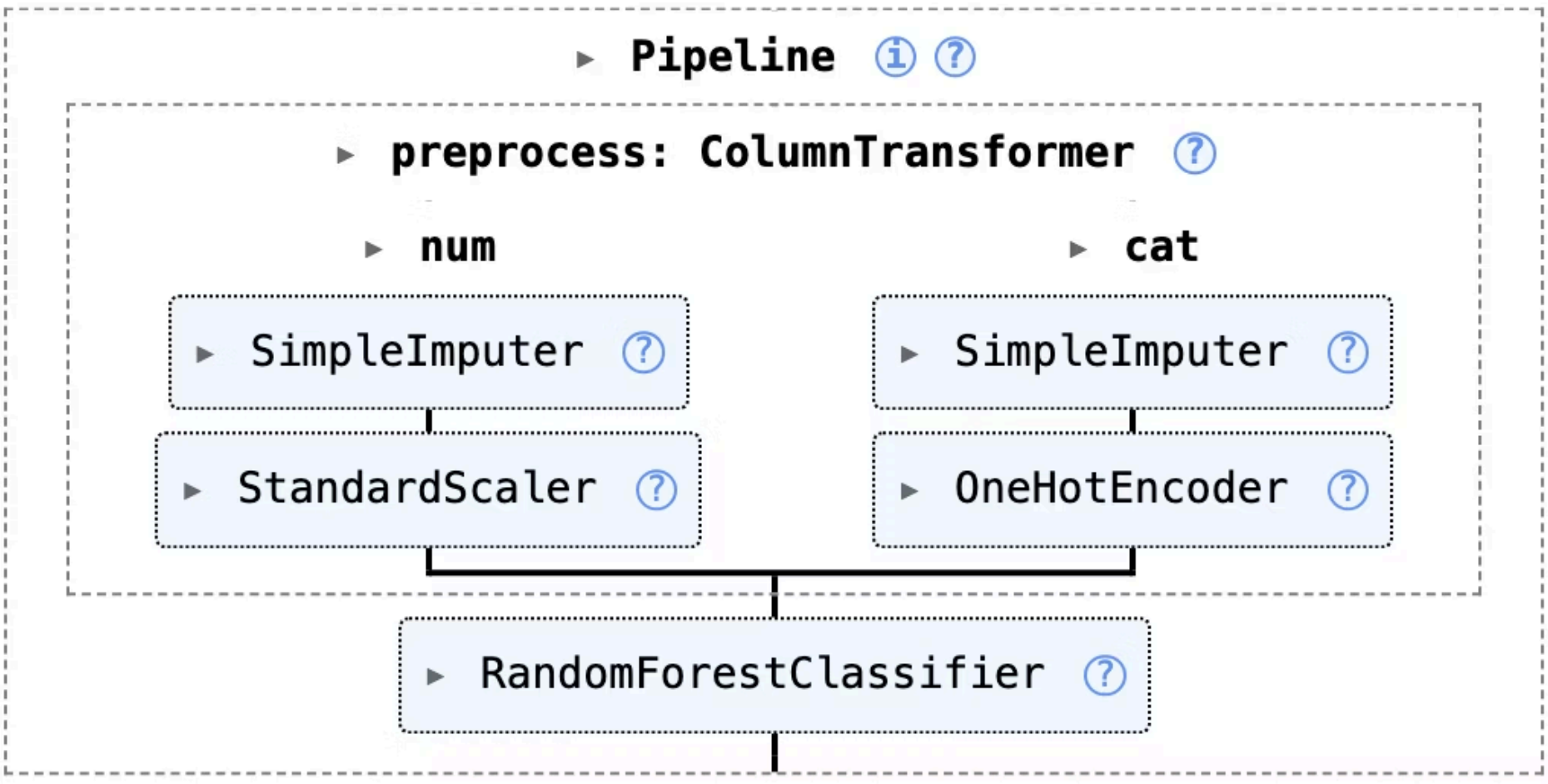
- Maintained by the **U.S. Consumer Product Safety Commission (CPSC)**
- Probability-based sample of ~**100 hospital emergency departments** nationwide
- Captures consumer product-related injuries presenting to EDs
- Study window: **2016–2024** — spanning pre- and post-COVID periods
- Records filtered using NEISS skiing and snowboarding product codes.

Variables Used in Analysis

Variable	Description
Age	Age of injured patient
Sex	Patient sex
Sport	Skiing or snowboarding indicator
Diagnosis	NEISS injury diagnosis code
Body_Part	Body region injured
Disposition	Patient outcome after ED visit
Year	Year of injury

Data Cleaning & Preparation

1. Filter and Subset
   a. Isolated skiing and snowboarding cases using NEISS product codes
2. Clean Categoricals
   a. Standardized race, diagnosis and body-part fields; resolved inconsistencies
3. Feature Engineering
   a. Created age groups; encoded categorical features; handled missing values
4. Modeling Pipeline
   a. Assembled reusable preprocessing pipeline for reproducible modeling



Predictive Modeling Strategy

Goal: Classify injury records as **pre-COVID** vs. **post-COVID** to detect pandemic-related shifts in injury patterns.

Model 1 — Logistic Regression

Interpretable baseline; establishes performance floor for comparison

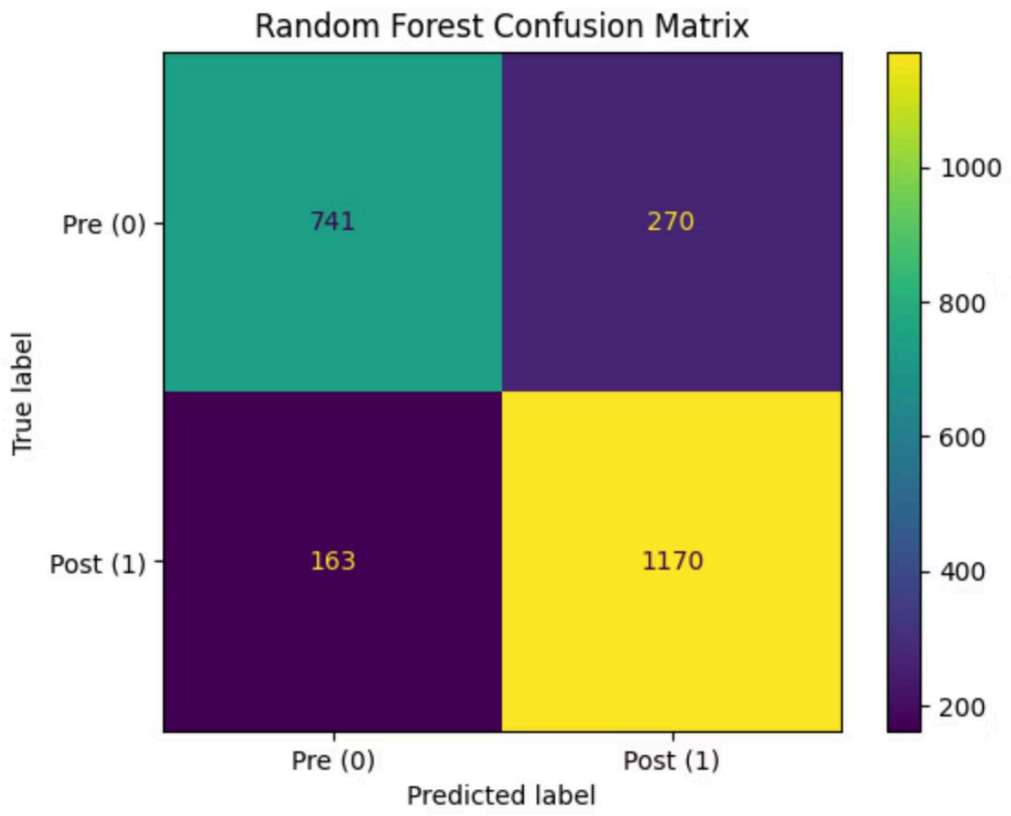
Model 2 — Random Forest

Ensemble method; captures nonlinear relationships and feature interactions

Evaluation Metrics

Accuracy · Precision · Recall · F1 Score · **ROC-AUC**

Model Performance Comparison



Logistic Regression

Accuracy: **0.690**

ROC-AUC: **0.748**

Random Forest

Accuracy: **0.815**

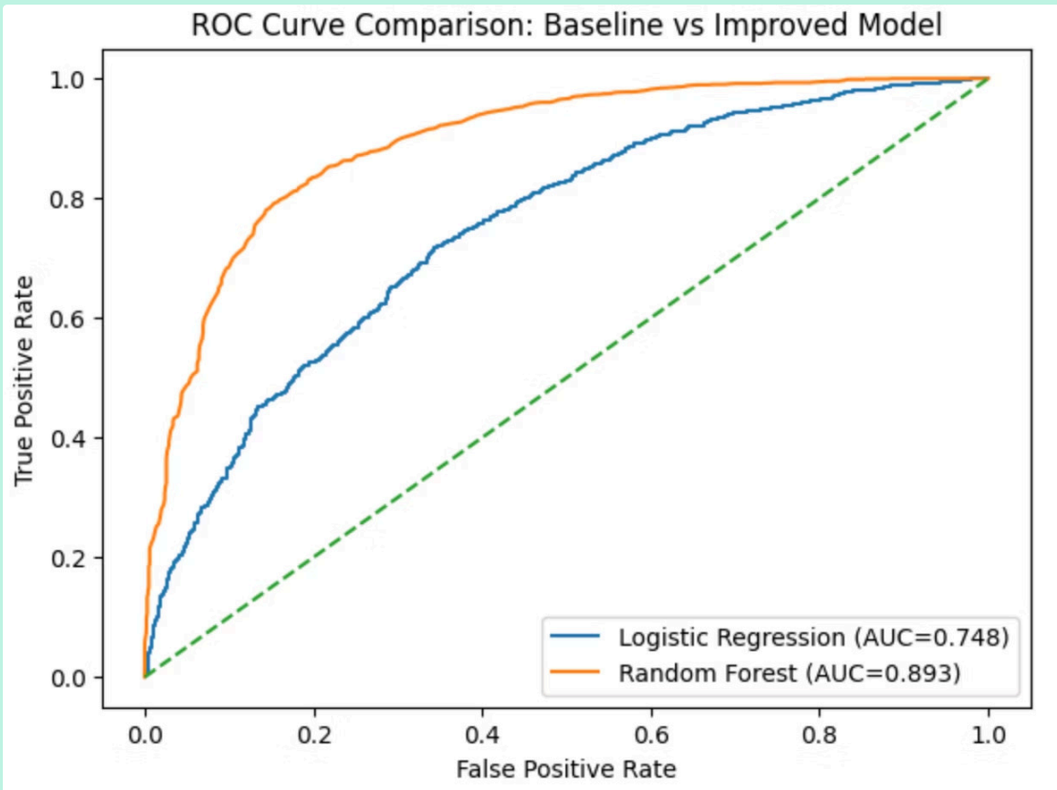
ROC-AUC: **0.893**

Key Takeaway

Random Forest improved accuracy by **+12.5 pp** and ROC-AUC by **+14.5 pp** over the logistic baseline.

Model Evaluation: ROC Curves

ROC curve plot for both models



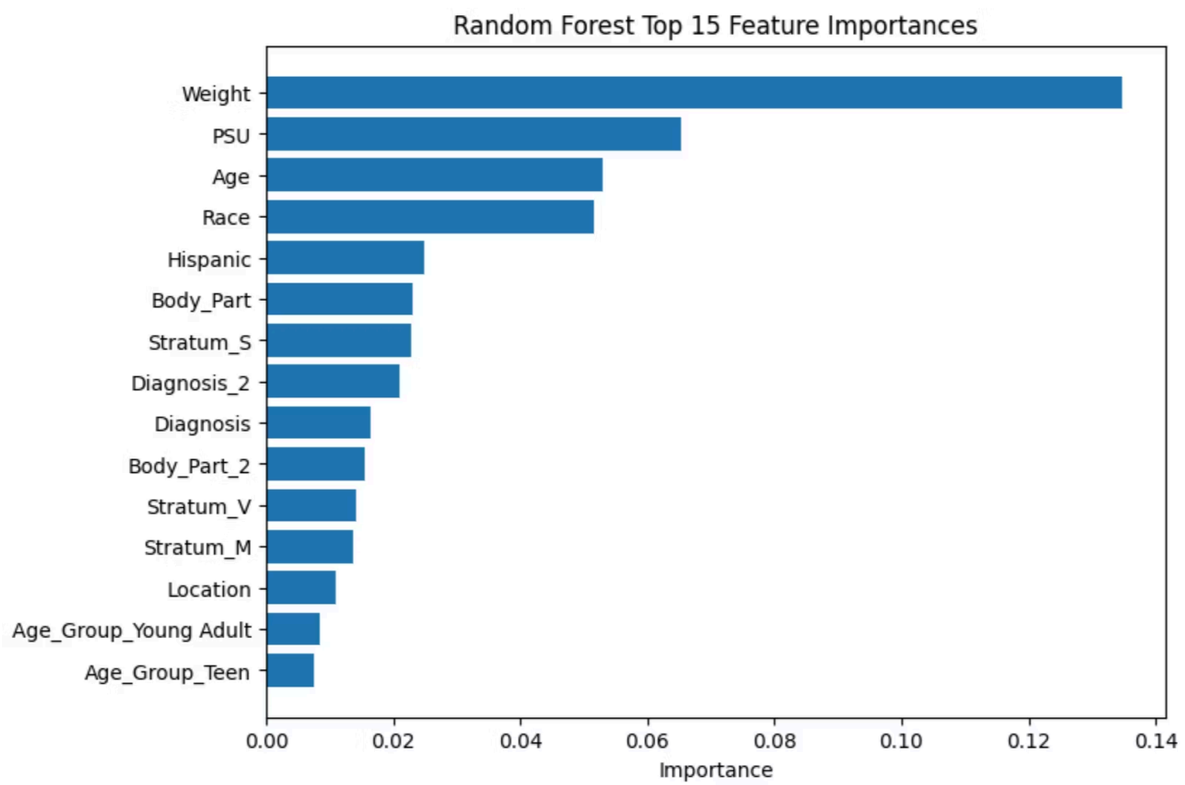
Logistic Regression (AUC = 0.748) vs. Random Forest (AUC = 0.893)

Interpreting ROC-AUC

- ROC curves plot **true positive rate** vs. **false positive rate** across classification thresholds
- Higher AUC reflects stronger discrimination between pre- and post-COVID injury records
- An AUC of 0.893 indicates strong discrimination between pre- and post-COVID injury records.

Result: Random Forest substantially outperformed Logistic Regression in discriminative ability.

Important Predictors: Feature Importance



Top Predictors

Demographic Factors

Age and race emerged as the strongest individual predictors in the model

Injury Characteristics

Body part injured, diagnosis type, and injury location contributed substantially

Survey Design Variables

NEISS weight and stratum variables captured sampling structure

⚠️ Feature importance reflects **model influence**, not causal effect on injury risk.

Key Findings & Future Directions

Demographic Patterns

Snow sport injuries follow clear demographic and temporal trends detectable through ED surveillance data

Machine Learning Detects Temporal Shift

Machine learning successfully identified pre- vs. post-COVID changes in injury patterns with strong discriminative performance

Model Performance

Random Forest (AUC 0.893) significantly outperformed the logistic baseline, validating the ensemble approach

Prevention Value

NEISS surveillance data can directly inform targeted injury prevention research and policy recommendations